Quantification: 548_hyp_HP_z08c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	25
interior lacuna count	20
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 20)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2518	1167.5	2379.27	px^2
major_axis_length_px	lacuna shape	77.693	71.1119	42.4097	px
minor_axis_length_px	lacuna shape	39.1383	30.447	21.1192	px
aspect_ratio	lacuna shape	2.1359	1.7159	1.0188	unitless
eccentricity	lacuna shape	0.8016	0.8109	0.1333	unitless
solidity	lacuna shape	0.7951	0.8106	0.0979	unitless
roots_count	headline	9.6	6.5	6.2357	count
ring_length_r30_px	headline	288.75	227.5	165.555	px
ring_length_r60_px	reported	513.4	416	287.6	px
owned_length_px	ownership-dependent	686.788	359.413	613.004	px
edge_count	ownership-dependent	34.05	17	29.452	count
mean_edge_length_px	ownership-dependent	19.7817	19.726	3.8376	px
perimeter_px	normalised	249.72	200.723	148.406	px
ring_area_r30_px2	normalised	7970.25	6784.5	4090.31	px^2
ring_area_r60_px2	normalised	16616.8	15515.5	8633.13	px^2
in_frame_fraction_r30	normalised	0.9864	1	0.0371	unitless
in_frame_fraction_r60	normalised	0.9505	1	0.0827	unitless
ring_density_r30	normalised	0.0370451	0.0376301	0.0101024	px^-1
ring_density_r60	normalised	0.0328011	0.0308998	0.0103793	px^-1
roots_per_100px_perimeter	normalised	3.9728	4.0373	1.1372	per 100 px
ring_attached_length_r30_px	network v2	253.45	192	165.885	px
ring_attached_length_r60_px	network v2	373.4	277.5	259.321	px
ring_length_w_r30_px	network v2	329.922	257.344	188.988	px
ring_length_w_r60_px	network v2	587.079	479.623	326.465	px
sholl_crossings_r10	network v2	7.05	5.5	4.2361	count
sholl_crossings_r20	network v2	7.6	7	4.4414	count
sholl_crossings_r30	network v2	6.7	5	4.6464	count
loop_count	loops, report only	1.5	0.5	1.9868	count
loop_edge_share	loops, report only	0.0971	0.0319	0.1256	share
ring_width_mean_r30_px	width	7.2533	7.4189	0.8138	px

Per field
----

measure	value	units
analysed_area_px2	982527	px^2
total_skeleton_length_px	27363	px
canalicular_length_density_per_px	0.0278496	px^-1
junction_count	546	count
junction_density_per_px2	0.00055571	px^-2
lacuna_count	25	count

lacunae_per_px2	2.544e-05	px^-2
skeleton_component_count	377	count
mean_component_length_px	72.5809	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0318588	px^-1
field_density_without_flagged_per_px	0.0278496	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	277	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0278496	px^-1
field_length_density_with_unattached_per_px	0.0281315	px^-1
loop_count	43	count
loop_edge_share	0.1049	share
edge_count	1458	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6474	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	27363	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1925	65.1309	45.5827	1.4289	0.7143	0.7606	0.0397	28.706	675.968	True
2	3961	122.826	46.3249	2.6514	0.9261	0.8431	0.4753	53.5506	452.253	False
3	547	38.4822	18.6369	2.0648	0.8749	0.9071	1.3303	28.7715	719.596	False
4	4287	134.045	52.6927	2.5439	0.9195	0.6794	-0.8737	66.8097	864.372	False
5	466	30.4448	24.0714	1.2648	0.6123	0.7373	-0.4028	72.367	793.723	False
6	3031	74.7141	59.7338	1.2508	0.6007	0.7786	-0.0729	198.233	105.492	False
7	1249	97.3283	20.9931	4.6362	0.9765	0.6665	0.811	244.751	234.965	False
8	833	42.3386	25.9256	1.6331	0.7906	0.8976	0.935	290.168	253.631	False
9	8164	134.063	85.066	1.576	0.7729	0.8013	-0.8691	359.846	342.936	False
10	5658	108.902	72.88	1.4943	0.7431	0.7761	-0.0527	409.015	685.903	False
11	1968	65.0915	47.7903	1.362	0.6789	0.7977	-1.2589	372.308	23.5991	True
12	429	26.825	21.927	1.2234	0.5761	0.8994	0.7907	406.583	41.8998	False
13	2190	104.344	39.1379	2.6661	0.927	0.7344	0.1875	474.681	15.2301	True
14	2786	87.8405	47.2353	1.8596	0.8431	0.8199	0.6054	469.408	431.352	False
15	4506	98.6377	61.8033	1.596	0.7794	0.8877	-0.0114	577.89	721.96	False
16	749	67.5097	15.6293	4.3194	0.9728	0.8397	0.4961	722.103	490.606	False
17	534	30.0222	23.0801	1.3008	0.6395	0.9287	0.8021	729.066	431.221	False
18	1086	59.0095	32.8062	1.7987	0.8312	0.7126	1.4627	783.261	210.357	False
19	4263	130.21	50.8324	2.5616	0.9207	0.6991	-1.4214	797.485	870.568	False
20	537	37.6344	28.0879	1.3399	0.6656	0.5582	-1.4625	836.14	208.279	False
21	1744	52.1215	44.4583	1.1724	0.522	0.9483	-1.1943	879.793	1002.11	True
22	6366	155.327	60.2608	2.5776	0.9217	0.7519	-1.2656	913.286	921.258	False
23	7862	142.695	80.4282	1.7742	0.826	0.8281	1.2173	982.477	693.681	True
24	458	30.6993	21.7842	1.4092	0.7046	0.8388	0.4457	956.432	222.284	False

25 450 47.0004 12.9942 3.617 0.961 0.8789 0.4432 967.353 249.747 False

Network per lacuna

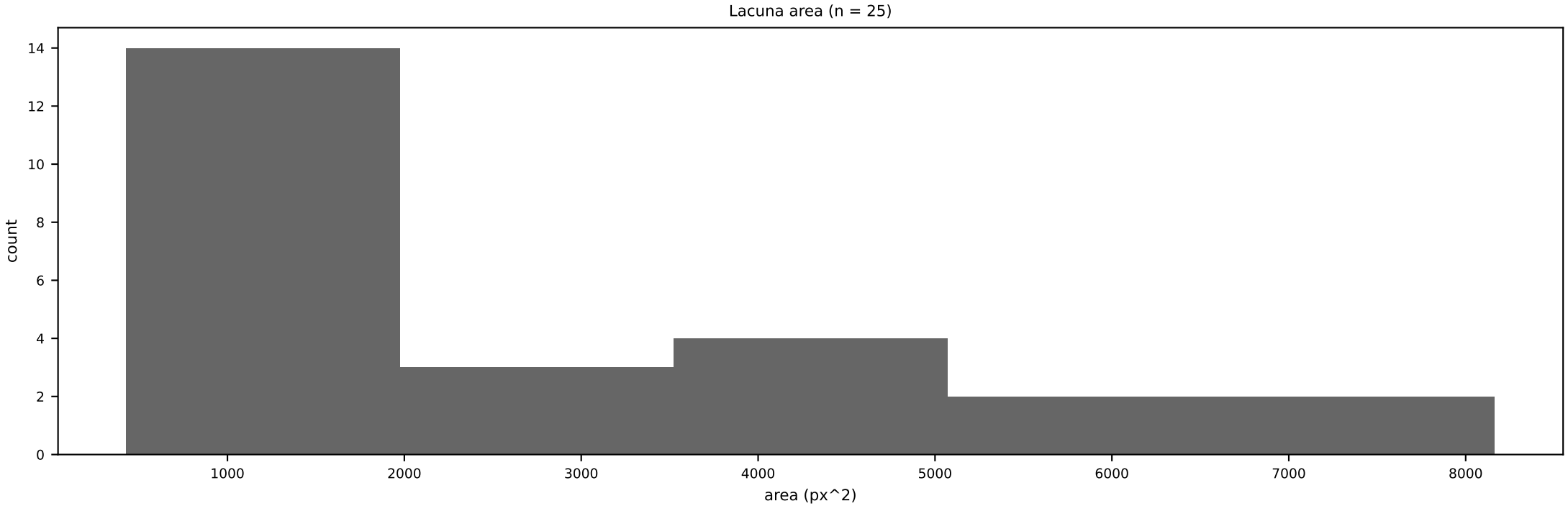
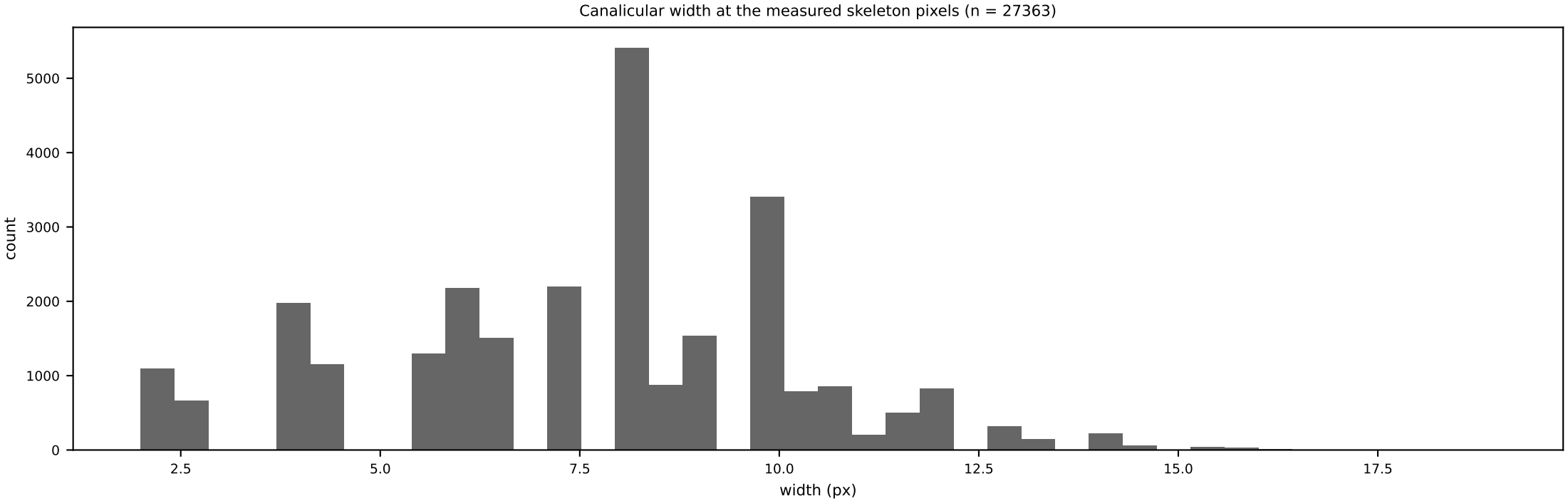
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	196	346	687.34	23	29.8844	6.7244
2	9	251	503	1112.97	68	16.3673	8.3083
3	6	178	223	243.409	14	17.3864	6.0163
4	18	428	689	1604.6	74	21.6838	7.4776
5	7	204	315	224.865	17	13.2274	6.7591
6	6	403	827	1056.68	47	22.4825	7.1285
7	10	278	427	273.794	17	16.1055	7.3603
8	5	137	226	197.167	10	19.7167	8.2717
9	25	686	1221	2131.41	108	19.7353	7.7058
10	20	427	776	952.051	50	19.041	7.6251
11	2	151	224	234.652	13	18.0501	6.6328
12	5	113	181	256.823	15	17.1216	7.4842
13	7	145	305	1441.67	63	22.8836	7.9314
14	6	373	723	1361.28	53	25.6846	8.1493
15	13	348	654	1562.55	72	21.7021	7.8628
16	5	194	405	445.031	15	29.6688	7.6226
17	2	80	223	17.0711	1	17.0711	5.8574
18	7	146	373	210.581	13	16.1985	6.9006
19	16	515	885	715.169	43	16.6318	6.9922
20	6	180	301	167.853	7	23.979	5.4322
21	0	8	50	0	0	0	9.2699
22	16	522	774	880.222	41	21.4688	8.3277
23	9	338	628	562.215	36	15.6171	7.3058
24	5	144	260	178.581	9	19.8423	6.9045
25	5	168	282	143.64	7	20.5199	6.8795

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	220.61	5180	11537	0.7389	0.6782	0.0378378	0.0299905	2.7197
2	327.706	10056	22309	0.8579	0.7712	0.0249602	0.0225469	2.7464
3	102.326	3114	3907	0.9041	0.7441	0.0571612	0.057077	5.8636
4	434.274	11650	22490	0.9794	0.8718	0.0367382	0.0306358	4.1448
5	117.225	3548	5322	1	1	0.0574972	0.0591883	5.9714
6	274.007	10299	23608	1	1	0.03913	0.0350305	2.1897
7	225.38	6965	13598	1	1	0.0399139	0.0314017	4.437
8	121.983	3984	7252	1	1	0.0343876	0.0311638	4.0989
9	522.985	15465	29728	1	1	0.0443582	0.0410724	4.7803
10	451.63	13578	29430	1	1	0.0314479	0.0263676	4.4284
11	224.368	4110	9102	0.697	0.6419	0.0367397	0.02461	0.8914
12	87.1543	2395	4224	0.996	0.8463	0.0471816	0.0428504	5.7369
13	356.534	5302	10449	0.6133	0.5772	0.0273482	0.0291894	1.9633
14	268.777	10180	22542	1	1	0.0366405	0.0320735	2.2323
15	328.534	11289	25601	1	1	0.0308265	0.0255459	3.957
16	155.681	6604	15865	1	1	0.0293761	0.0255279	3.2117
17	95.6396	4787	11479	1	1	0.0167119	0.0194268	2.0912
18	176.066	6367	15166	1	1	0.0229307	0.0245945	3.9758
19	441.789	13369	27214	1	1	0.0385219	0.03252	3.6216
20	157.652	5421	11799	1	1	0.0332042	0.0255106	3.8059
21	173.397	1958	3433	0.6626	0.6178	0.0040858	0.0145645	0
22	478.902	12272	22431	0.9903	0.9107	0.0425359	0.0345058	3.341
23	502.902	11117	23341	0.7175	0.6712	0.0304039	0.0269055	1.7896

24	112.811	3737	8995	1	0.9788	0.0385336	0.0289049	4.4322	
25	113.882	4325	9376	1	0.887	0.0388439	0.0300768	4.3905	

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	173	230	221.995	398.676	5	5	4
2	206	281	282.794	578.63	6	6	4
3	161	191	201.681	253.723	3	4	2
4	359	581	482.061	786.21	10	8	11
5	178	219	231.894	370.747	3	5	6
6	336	704	457.747	935.151	8	14	14
7	220	286	315.25	478.375	6	11	5
8	105	164	163.095	261.522	2	5	4
9	663	997	787.311	1390.21	16	20	12
10	385	560	488.345	891.607	11	9	11
11	136	185	169.125	254.38	4	2	4
12	109	177	131.225	207.924	3	4	2
13	145	256	153.598	339.936	3	5	3
14	348	560	432.718	825.767	14	8	10
15	339	562	396.049	741.713	10	8	13
16	153	274	230.451	480.872	4	8	5
17	54	54	90.1543	253.693	5	2	2
18	132	192	165.61	430.103	5	5	3
19	487	594	587.316	1007.48	12	12	6
20	133	168	202.581	339.22	4	4	1
21	0	0	8.2426	58.4264	0	0	1
22	502	688	596.144	885.293	12	12	16
23	309	422	388.019	704.813	8	5	7
24	89	106	167.924	303.534	3	4	4
25	110	110	188.095	319.806	4	3	3

548_hyp_HP_z08c1-2: distributions, pre-validation, px